

Kaustav Mukherjee

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EDUCATION

Carnegie Mellon University (CMU)

Master of Science in Computer Vision (GPA 4.22/4.33)

Pittsburgh, PA

Aug. 2024 – Dec. 2025

National University of Singapore (NUS)

Bachelor of Engineering in Mechanical Engineering (GPA 4.64/5.00)

Singapore, Singapore

Aug. 2020 – May 2024

EXPERIENCE

AI Engineer

Jan. 2026 – Present

Kaliber Labs

San Francisco, CA

- Led a 4-person research team developing a clinical multimodal conversational AI MVP, translating clinic feedback into product requirements, establishing a continual data labeling and training loop, and delivering investor demos
- Built a real-time multi-view 3D reconstruction pipeline from surgical video using TSDF fusion, driving a 2-day showcase resulting in an active acquisition process with an F500 medtech company; published at ECCV DCA-MI
- Researched latent world models, implementing Mamba-based geometry conditioning that improved navigation success rate in dynamic 3D scenes by 36%, and developing methods to physically ground latent spaces

AI Engineer Intern

May 2025 – Aug. 2025

Kaliber Labs

San Francisco, CA

- Architected a modular robotic navigation system with ROS2 and Nav 2, integrating 6+ CV models with C++ behavior trees via shared memory and multithreading, and deployed on the LG Q9 robot and NVIDIA Isaac Sim
- Reduced per-frame edge processing latency from 300ms to 150ms across multiple CV inference tasks
- Led technical integration with software, robotics, MLE and research teams, presenting solutions to 10+ investors

Software Engineering Intern

Jan. 2023 – Jul. 2024

Kaliber Labs

San Francisco, CA

- Accelerated convergence of a surgical camera pose estimation algorithm by 10 times using particle swarm optimization (PSO) to match semantic segmentation masks between the scope view and a digital twin
- Utilized NeRFs for real-time camera registration, enabling geodesic measurements with 90% accuracy
- Fine-tuned stable diffusion to generate synthetic training data, expanding surgical tool datasets by up to 5 times
- Employed artificial data augmentation to train multiple semantic segmentation models to above 97% accuracy
- Optimized CV model inference pipelines via ONNX and quantization, reducing memory footprint by 40%

PUBLICATIONS

Bychkov, D., Shah, R., Mukherjee, K., So, M. "Surface Coverage Completeness for Quality-Assured Arthroscopic 3D Reconstruction" *Proceedings of the ECCV Workshop on DCA-MI, 2026*

Mukherjee, K. et al. "Geometry-Conditioned Latent World Models with Recurrent Adaptation in Dynamic Environments" *Submitted to ICLR 2027 (Under Review)*

Zhu, P., Penachio, S., Mukherjee, K., Jonelagadda, A. "Spectral-Target Physical Latent Structuring for JEPA-Style World Models" *Submitted to ICLR 2027 (Under Review)*

PROJECTS

Bundle Adjustment for 360° Pose Correction | Gaussian Splatting, SfM, SLAM

Aug. 2025 – Nov. 2025

- Implemented a rig-based Structure from Motion (SfM) pipeline with bundle adjustment for Stella-VSLAM outputs from 360° videos, improving Gaussian Splatting SSIM by 10.5% on a self-collected hour-long outdoor dataset

Drone Perception System at CMU Airlab | Python, OpenCV, ROS, Computer Vision

Aug. 2024 – Nov. 2024

- Designed a depth-aware perception system for a drone with a 2-DOF manipulator using ROS and RGBD cameras

SKILLS

Competencies: NeRF, Gaussian Splatting, World Models, 3D Reconstruction, Computer Vision, SLAM, SfM, Diffusion, Reinforcement Learning, LLM Fine-tuning (RLHF, DPO), Context Engineering, Model Quantization

Frameworks: PyTorch, Tensorflow, NumPy, OpenCV, ROS2

Languages and Tools: Python, C++, JavaScript, Docker, Redis, NATS, ONNX, AWS, Linux, Git